

Acknowledgements

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Abstract

Russia’s full-scale invasion of Ukraine in February 2022 triggered a sharp energy price shock in Europe, with potentially heterogeneous effects across industrial sectors. This paper studies how the war and the associated energy crisis affected German manufacturing, focusing on differences by sectoral energy intensity. Using a monthly panel of 4-digit manufacturing industries (WZ08) from 2015 to 2024, we estimate difference-in-differences models with continuous treatment intensity, interacting the post-war period with pre-determined sectoral energy shares. Findings reveal that energy-intensive sectors experienced significantly larger declines in industrial production after March 2022, while employment and hours responded more weakly, consistent with labour hoarding and adjustment friction. Dynamic event-study analyses reveal that pre-trends differ across sectors but that, once sector-specific linear trends are accounted for, the post-war production decline remains pronounced and persistent. Additional evidence suggests that energy-intensive sectors co-moved more strongly with the collapse in German exports to Russia, whereas general foreign exposure played a limited role. Aggregate macroeconomic indicators such as GDP and investment show little response, highlighting the importance of sector-level analysis. Overall, the results underscore that energy price shocks primarily operate through cost exposure, generating substantial heterogeneity within manufacturing rather than broad-based deindustrialisation.

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List of Abbreviations and Symbols

CEPR	Centre for Economic Policy Research
COVID-19	Coronavirus Disease 2019
DGAP	Deutsche Gesellschaft für Auswärtige Politik (German Council on Foreign Relations)
GFCF	Gross Fixed Capital Formation
GDP	Gross Domestic Product
GCEE	German Council of Economic Experts
IAB	Institute for Employment Research (Germany)
NBER	National Bureau of Economic Research
ECB	European Central Bank
WDI	World Development Indicators
WZ08	German Classification of Economic Activities 2008
ZEW	Zentrum für Europäische Wirtschaftsforschung
WISTA	Wirtschaft und Statistik (German statistical journal)
SD	Standard Deviation

1 Introduction

Russia’s full-scale invasion of Ukraine in February 2022 unleashed an unprecedented energy price shock across Europe, disrupting trade flows, upending gas supply chains, and triggering concerns about deindustrialising countries heavily dependent on imported fossil fuels. Germany—whose growth model combines a large manufacturing base with high reliance on imported fossil fuels and historical dependence on Russian gas—was among the most affected economies (German Council of Economic Experts, 2022). According to the Federal Statistical Office, the production index for industry is a volume index that measures price-adjusted production relative to a base year and is constructed as a weighted average of branch-specific indices (Destatis, 2023). Energy prices in Germany sharply increased by mid-2022 compared to the previous year, particularly at the wholesale and industrial level. Official statistics show that natural gas prices in February 2022 were nearly three-to-four times higher than a year earlier for imports and industrial users, while electricity prices for industrial consumers also increased significantly over the same period, and energy-intensive sectors such as basic metals, glass, ceramics, and chemical products were disproportionately affected. Policy debates centred on whether the shock would trigger a wave of deindustrialisation or instead induce sector-specific adjustment and reallocation. Previous research emphasises that energy supply shocks tend to lower industrial output, especially for energy-intensive sectors, while imports of the same goods increase (Anderton and Carter, 2001; Bachmann et al., 2024). In parallel, the collapse of German exports to Russia after the invasion posed an additional trade shock. The rest of the introduction provides context and literature review.

1.1 Economic context and literature

Energy prices in Germany had already risen significantly at the end of 2021, but “the situation was made drastically worse by Russia’s attack on Ukraine, which caused energy prices to soar” (?). To monitor the shock, the Federal Statistical Office responded by publishing a new production index for energy-intensive industrial branches from June 2022 onwards. Using cross-country data, Chiacchio et al. (2023) show that euro-area production of high energy-intensive goods fell markedly from mid-2022, while imports of those goods increased; producers substituted expensive domestic production with cheaper imports. Bachmann et al. (2024) use a multi-sector open economy model to quantify the macroeconomic costs of a stop to Russian energy imports; they emphasise that aggregate GDP losses are likely manageable (0.5–3%), but energy-intensive sectors experience more severe effects. Hutter et al. (2023) document how German exports to Russia collapsed after the invasion and were partly redirected to domestic and eurozone markets with little aggregate production effect. Firm-level surveys during the crisis show that high energy prices triggered a range of short-term responses—including substituting machinery, changing suppliers, adopting energy management systems, and relocating energy-intensive activities—and that firms with high energy intensity suffered negative sales growth. Structural analysis underscores that rising energy prices accelerate industrial restructuring but do not necessarily lead to broad deindustrialisation. The German Council of Economic Experts notes that although energy prices are expected to fall gradually, they will remain above pre-crisis levels; energy-intensive industries facing competition from non-European producers are particularly vulnerable, yet widespread deindustrialisation is unlikely. A CEPR VoxEU column argues that cost increases are

unequally distributed across sectors—for example, chemicals, food products, basic metals and non-metallic minerals account for most natural gas usage—and stresses that adaptation margins (innovation, fuel switching, own electricity generation) can buffer cost shocks. Large industrial consumers in Germany reduced gas usage by about 15% in 2022 while maintaining output. These perspectives suggest that heterogeneity across industries, rather than aggregate averages, is key to understanding the crisis. International evidence also shows that energy price differences shape the geography of manufacturing and can trigger cross-border reallocation: [Manderson and Kneller \(2020\)](#) exploit the US shale gas revolution as a quasi-natural experiment and conclude that energy-intensive UK multinationals established new affiliates in the United States and closed energy-intensive plants in the UK because the energy price gap made US production cheaper. [Kahn and Mansur \(2010\)](#) document that energy-intensive industries concentrate in U.S. counties with low electricity prices, while labour-intensive sectors avoid pro-union counties, and pollution-intensive industries locate in areas with lax environmental regulation. During the current crisis, high energy prices have renewed these relocation incentives; a DGAP memo reports that Germany has already shifted parts of its most energy-intensive industrial production abroad, resulting in carbon leakage.

1.2 Contribution of this paper

This seminar paper assembles a monthly panel of 283 German manufacturing industries at the 4-digit WZ08 level (divisions 10–33¹) from January 2015 through December 2024 and estimates how the post-March-2022 energy crisis differentially affected sectors with varying energy intensity. The approach builds on difference-in-differences designs with continuous treatment intensity and dynamic event studies. The focus is on production (industrial production index), employment and hours worked, and we link sectoral outcomes to trade conditions and export-reallocation dynamics via monthly Germany–Russia trade data. By combining detailed sectoral data with cross-sectional energy-intensity measures, we quantify the heterogeneous impact of the energy shock and provide evidence on how sector characteristics mediate adaptation. The analysis complements macro models by emphasising within-economy distributional effects and enriches the export-redirection framework of [Hutter et al. \(2023\)](#). We also report simple macro regressions for Germany’s GDP, gross fixed capital formation and capital stock to contrast aggregate and sectoral responses.

Previewing the findings: In a two-way fixed-effects model, we estimate that a one-standard-deviation (1 SD) higher energy intensity is associated with an additional 4.1% decline in industrial production in the post-March-2022 period, while employment and hours decline by around 1.7% per SD. Event-study estimates reveal persistent negative effects and motivate the inclusion of sector-specific linear trends to address differential pre-trends. Macro regressions show only a modest and statistically insignificant impact of the war dummy on aggregate log GDP and investment, highlighting that heterogeneity is masked at the aggregate level. Interactions with a time-varying Russia trade shock suggest that the collapse of exports to Russia co-moved with larger output losses in energy-intensive sectors. The results support the view that energy price shocks create distributional effects across sectors rather than uniform downturns.

¹Divisions 10–33 of WZ 2008 constitute the full manufacturing sector. Within these divisions, the dataset contains 283 distinct four-digit industries.

2 Data

2.1 Sectoral outcomes: production, turnover and foreign exposure

The baseline panel comprises monthly observations for 283 four-digit manufacturing industries (divisions 10–33 of WZ 2008) between January 2015 and December 2024. The dependent variables are as follows:

- **Industrial production index** $\ln_prod$: the natural logarithm of the Destatis production index at the 4-digit level. Destatis describes this index as a volume index of price-adjusted production, constructed as a fixed-base weighted average of branch-specific indices (weights reflect gross value added); the index provides a high-frequency measure of real output.
- **Turnover measures**: monthly turnover is available separately for domestic, foreign, euro-area and non-euro-area sales. We compute the foreign turnover share as foreign turnover divided by total turnover and the non-euro foreign share as non-euro turnover divided by foreign turnover. These shares serve as proxies for each sector’s external orientation and are used both descriptively and as interaction variables.

Descriptively, the distribution of the production index shows pronounced heterogeneity: heavy industries such as industrial gases, basic metals, glass and ceramics display large swings during 2022–2023, while less energy-intensive sectors (e.g. clothing repair, tobacco) remain comparatively stable. Turnover data reveal that some high-energy-intensity sectors increased the share of non-euro-area sales during the crisis, hinting at export reallocation. To link sector outcomes to Russia-related trade shocks, we merge monthly Germany–Russia export and import series from Eurostat. Because the merge produced a “many-to-many” warning due to duplicate dates, we group the trade series by month and sum values to ensure one observation per month before joining. We then define a Russia shock series as the deviation of log exports to Russia from their pre-war mean (negative during the war period).

2.2 Energy intensity and exposure

We measure cross-sector energy intensity using the energy consumption share of gross production value (a Destatis cost-structure statistic). This variable, denoted $energy_share_i$, reflects each sector’s direct energy cost share and is time-invariant over our sample. To aid interpretation, we standardise this measure:

$$energy_std,i = \frac{energy_share_i - \overline{energy_share}}{sd(energy_share)}, \quad (1)$$

where $\overline{energy_share}$ and $sd(energy_share)$ are the cross-sectional mean and standard deviation of the raw energy share. Sectors such as industrial gases, glass, cement, lime/gypsum, ceramics and basic metals lie in the right tail of the energy-intensity distribution, while repair services, tobacco, spirits and some electronics are in the left tail. We compute pre-war averages of the foreign and non-euro turnover shares to construct

sector-specific exposure measures $exp_foreign,i$ and $exp_noneuro,i$. These proxies reflect general openness rather than Russia-specific dependence, similar to the exposure used by [Hutter et al. \(2023\)](#), but allow us to test whether sectors that are more externally oriented respond differently to the trade shock.

2.3 Employment and hours

Monthly employment and hours worked come from the GENESIS table 42111-0004. After reshaping and matching to WZ codes, we obtain roughly 27,000 sector-month observations for employment and hours. We use log employment $\ln_emp$ and log hours worked $\ln_hours$ as dependent variables. These labour-market series exhibit less volatility than production and adjust more slowly around the crisis, consistent with labour-hoarding behaviour and institutional frictions in Germany’s labour market (e.g., short-time work schemes and working-time accounts). Figure [1](#) illustrates labour-market responses by energy intensity with sector-specific trends.

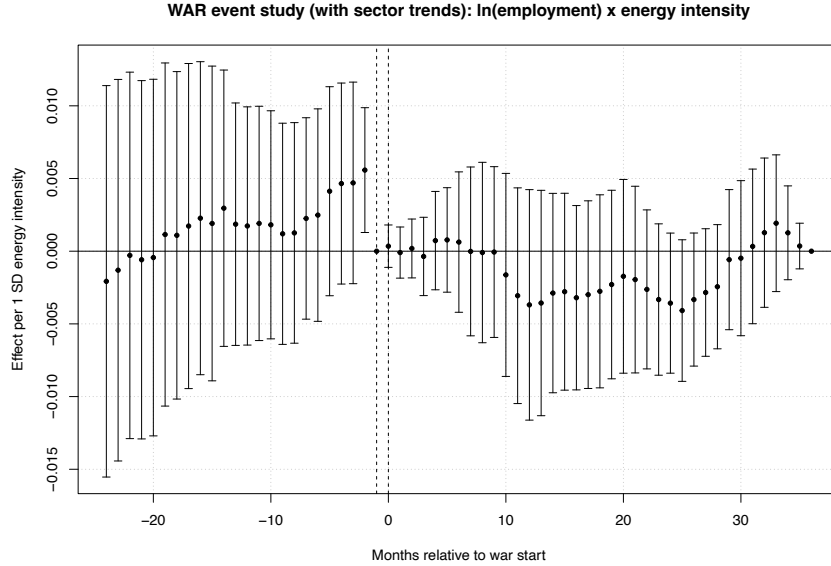


Figure 1: Labour-market responses to the war by energy intensity (with sector trends). Notes: event-study estimates of the interaction between energy intensity and the war indicator for log employment; the shaded area depicts 90% confidence intervals.

2.4 Macro context

For macroeconomic context, we extract annual time series for Germany (2015–2025) from the World Bank WDI and national accounts. The average annual GDP growth rate falls from 1.22% pre-war to 0.15% in the war period, while log GDP remains essentially flat, reflecting stagnation rather than collapse. Investment and capital stock measures display modest changes. To provide context, Figure [3](#) plots annual real GDP growth for Germany with a vertical line marking March 2022, the war start. Figure [4](#) shows the share of gross fixed capital formation (GFCF) in GDP. Figure [5](#) plots monthly German exports to Russia, highlighting the trade shock.

3 Empirical Strategy

3.1 Baseline difference-in-differences with continuous treatment

The empirical strategy builds on the difference-in-differences (DiD) framework with continuous treatment intensity (Angrist and Pischke, 2009; Callaway and Sant’Anna, 2021) and dynamic event-study designs (Schmidheiny and Siegloch, 2019). To quantify how the energy shock affected sectors with different energy intensity, we estimate a DiD model with continuous treatment intensity. Let y_{it} denote the log outcome of sector i in month t (production, employment or hours). Define War_t as an indicator equal to 1 for months from March 2022 onwards and 0 otherwise, and let Energy_i be the standardised energy share from equation (1). The baseline specification is

$$\ln(y_{it}) = \alpha_i + \delta_t + \beta \text{War}_t \times \text{Energy}_i + \epsilon_{it}, \quad (2)$$

where α_i and δ_t are sector and month fixed effects. The interaction coefficient β captures whether more energy-intensive sectors experienced larger changes after the war relative to less intensive sectors. Standard errors are clustered at the sector level to allow for arbitrary serial correlation within industries.

3.2 Dynamic event-study specification

Following the event-study literature—which interprets DiD coefficients as weighted averages of dynamic causal effects and stresses the importance of pre-trend diagnostics—we estimate a dynamic event study with interactions between month-relative dummies and energy intensity:

$$\ln(y_{it}) = \alpha_i + \delta_t + \sum_{k \neq -1} \beta_k \mathbf{1}(t - t_0 = k) \times \text{Energy}_i + \epsilon_{it}, \quad (3)$$

where t_0 denotes March 2022 and k indexes months relative to the shock; the coefficient on $k = -1$ serves as the omitted baseline. This specification allows us to visualise pre-war and post-war dynamics. The identifying assumption is that, absent the shock, trends in high- and low-energy sectors would have evolved in parallel after controlling for sector and month fixed effects.

3.3 Pre-trend diagnostics and sector-specific trends

Because treatment intensity is continuous, the standard parallel-trends assumption requires that the differential trends across energy intensity are flat before the shock. We conduct joint Wald tests on the pre-war coefficients in the dynamic specification and find that the null of zero pre-trends is rejected for production and employment, indicating that energy-intensive sectors were already diverging before the war. To address this, we estimate event-study specifications with sector-specific linear trends:

$$\ln(y_{it}) = \alpha_i + \delta_t + \delta_i t + \sum_{k \neq -1} \beta_k \mathbf{1}(t - t_0 = k) \times \text{Energy}_i + \epsilon_{it}, \quad (4)$$

where $\delta_i t$ allows each sector to have its own linear time trend. This is a common robustness strategy when the treatment intensity correlates with long-run structural decline (e.g.,

heavy industry shrinking relative to light manufacturing). The identifying assumption becomes that, conditional on sector trends, deviations from trend are comparable across the energy-intensity distribution. We plot event-study coefficients with and without sector trends in Figure 2 for production, employment and hours.

3.4 Trade-shock interactions

To examine whether the collapse of trade with Russia co-moved with sector outcomes, we interact a time-varying Russia trade shock ru_shock_t with sector characteristics. Specifically, we estimate

$$\ln y_{it} = \alpha_i + \delta_t + \delta_i t + \theta_1 (ru_shock_t \times exp_foreign, i) + \theta_2 (ru_shock_t \times Energy_i) + u_{it}, \quad (5)$$

where $exp_foreign, i$ is the pre-war foreign turnover share. Because ru_shock_t is negative during the war period, a positive θ_2 implies that more energy-intensive sectors suffer larger declines when exports to Russia collapse. This specification is inspired by Hutter et al. (2023) and provides suggestive evidence on how trade and energy shocks interact.

3.5 Parallel trends and identification

We assess the parallel-trends assumption using joint Wald tests on the pre-war event-study coefficients. In the baseline specification without sector-specific trends, these tests reject the null of zero pre-trends, indicating that more energy-intensive industries were already on different trajectories prior to 2022. Therefore, we adopt the specification with sector-specific linear trends as our preferred design. In this trend-adjusted specification, the joint pre-trend tests no longer reject, while post-war coefficients remain jointly significant. Formal test results are reported in the appendix.

4 Results

4.1 Main difference-in-differences estimates

Table 1 reports estimates of the baseline specification (2) for production, employment and hours. A one-standard-deviation increase in energy intensity is associated with an additional 4.08% decline in log production after March 2022 ($\beta = -0.0408$, $p < 0.001$). Converting from log points, this corresponds to roughly $100 \times (\exp(-0.0408) - 1) \approx -4.0\%$. For employment and hours, the estimated elasticities are around -1.7% and are only marginally significant at the 10% level.

Table 1: War exposure and energy intensity: baseline fixed-effects estimates

	Dependent variable		
	ln(Production)	ln(Employment)	ln(Hours)
War $\times$ Energy intensity	−0.0408*** (0.0109)	−0.0169 (0.0097)	−0.0173 (0.0097)
Sector FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	26 961	27 133	27 133
<i>Notes:</i> Each column reports OLS estimates of equation (2). Standard errors (in parentheses) are clustered by sector. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.			

The estimates confirm that the energy shock’s impact is heterogeneous: energy-intensive sectors reduce output substantially more than energy-light sectors. Importantly, the effect on employment and hours is smaller than on production. This gap is economically meaningful: firms often adjust output along margins other than labour—such as capacity utilisation, intermediate inputs, inventories or product mix—before reducing headcount. Labour markets in Germany are characterised by short-time work schemes, working-time accounts and collective agreements that facilitate hours adjustments while preserving employment. Firm-level surveys during the energy crisis document a menu of responses—substituting machinery, changing suppliers, adopting energy management systems, relocating activities and replacing fossil energy sources—which can mitigate immediate employment losses. In this sense, the “small” employment effect is evidence of labour smoothing rather than a statistical failure.

4.2 Dynamic event-study results

The dynamic event-study reveals the timing and persistence of differential effects. Without sector trends, the coefficients for leads ($k = -24$ to -2) show significant negative pre-trends for production and employment, implying that energy-intensive sectors were already declining relative to others before the war. Post-shock coefficients remain negative and large through 2023 and 2024, indicating persistent relative declines. When sector-specific linear trends are included, the pre-trend coefficients become statistically insignificant and centred around zero, while the post-shock coefficients remain negative and sizable for production. This pattern suggests that the war-period decline is not merely a continuation of pre-existing trends but represents an additional shock. For employment and hours, the post-shock coefficients are smaller and often not distinguishable from zero once trends are included, reinforcing the interpretation of labour smoothing. Figure 2 illustrates dynamic effects with sector trends for production.

4.3 Trade-shock interactions

Table 2 reports estimates from models that interact the time-varying Russia trade shock with sector characteristics, controlling for sector fixed effects, time fixed effects and sector-specific linear trends. The coefficient θ_2 on the interaction between the Russia trade shock and sectoral energy intensity is negative and statistically significant for all outcomes. For

production, the estimated coefficient is about -0.0037 , while for employment and hours the corresponding estimates are around -0.0014 and -0.0015 , respectively. Because the Russia trade shock is defined so that lower values correspond to more adverse trade conditions, a negative θ_2 implies that energy-intensive sectors experience larger declines when trade with Russia deteriorates. Although the magnitudes of these effects are smaller relative to the main war–energy interaction, they indicate that trade disruptions amplify the adverse impact of the energy shock rather than constituting an independent primary channel. By contrast, the coefficient on the interaction between the Russia trade shock and a sector’s generic foreign turnover share is close to zero and statistically insignificant across all outcomes, suggesting that overall export openness does not mediate the Russia trade shock. This finding aligns with [Hutter et al. \(2023\)](#), who show that German exports to Russia were largely redirected toward other destinations and therefore did not translate mechanically into aggregate production losses.

Table 2: Russia trade shock, energy intensity, and sectoral outcomes

	Dependent variable		
	ln(Production)	ln(Employment)	ln(Hours)
Russia shock $\times$ Energy intensity	-0.0037^* (0.0010)	-0.0014^* (0.0007)	-0.0015^* (0.0007)
Russia shock $\times$ Foreign export exposure	0.0024 (0.0047)	0.0001 (0.0037)	-0.0026 (0.0039)
Sector FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Sector trends	Yes	Yes	Yes
Observations	26 445	26 733	26 733

Notes: Standard errors clustered at the sector level. $*p < 0.05$.

4.4 Macro regressions

We place the sectoral results in context by estimating simple macro regressions of log real GDP, log gross fixed capital formation (GFCF) and log capital stock on a war dummy (2022–2025) and a COVID-19 dummy (2020–2021). With only ten annual observations, these results are illustrative. The war dummy is small and statistically insignificant for GDP (coefficient ≈ 0.030 , $p \approx 0.23$) and negative but insignificant for GFCF (≈ -0.022 , $p \approx 0.55$). For the capital stock, the war dummy is positive but insignificant, while the COVID dummy is significant and positive, reflecting the strong rebound after the pandemic. These findings indicate that at the macro level, Germany’s output and investment did not collapse during the energy crisis; instead, sectoral heterogeneity and reallocation mask aggregate stability. This pattern resonates with the literature: ? emphasise that a stop of Russian energy imports would have limited GDP effects overall but more pronounced sectoral consequences.

5 Discussion

5.1 Mechanisms and interpretation

The central mechanism behind our findings is the cost channel. The WISTA article on energy-intensive industrial branches notes that enterprises in these sectors require large amounts of energy to produce their goods, and high energy prices increase the risk of limiting economic activity. When energy prices surge, marginal costs rise more in energy-intensive sectors, inducing larger reductions in output. The estimated production elasticity of about 4% per SD aligns with this mechanism. Importantly, price changes alone do not dictate outcomes; substitution possibilities, input–output linkages and export opportunities also matter. Evidence from Fontagné et al. (2024), Freeman et al. (1997), and von Graevenitz and Rottner (2022) shows that energy price shocks lead to significant adjustments in production and trade patterns, but firms adapt through investment in energy-efficient technologies, relocation of production, and substitution toward imported inputs. Our interaction results with the Russia trade shock reinforce the notion that trade and energy shocks operate through complementary channels.

5.2 Heterogeneity and structural change

The crisis highlights the importance of heterogeneity. Energy-intensive sectors—such as basic metals, chemicals, glass and ceramics—account for a large share of industrial gas usage and thus face greater cost pressure. Yet the CEPR column notes that even within these sectors, costs are concentrated in a few product groups and that firms have multiple margins for adaptation—innovation, fuel switching, own electricity generation and process adjustments. Large industrial consumers reduced gas usage by around 15% in 2022 without cutting output. The GCEE argues that energy-intensive industries may relocate some production abroad, but broad-based deindustrialisation is unlikely; this underscores the role of targeted policies that facilitate structural change rather than preserving existing structures. Our results are consistent with these narratives: concentrated in energy-intensive sectors, the production decline persists over the first two years after the war. Employment declines are modest, suggesting that firms smooth labour adjustments and exploit other margins. Interactions with Russia trade shocks indicate that trade disruptions and energy shocks are complementary rather than separate channels. Overall, the energy crisis appears to accelerate structural change: sectors with high energy intensity and limited substitution opportunities face contraction, while other sectors expand or adapt.

5.3 Limitations and robustness

Several limitations merit discussion. First, the measure of energy intensity is time-invariant and may not capture changes in energy efficiency during the sample period; firms investing in energy-saving technologies could alter their cost structure, attenuating measured exposure. Second, the Russia trade shock is measured at the aggregate level rather than by sector, so the interaction results are suggestive rather than causal. Third, dynamic specifications with sector trends rely on the assumption that unobserved non-linear trends are negligible; alternative identification strategies—such as instrumenting energy prices or using variation in gas price pass-through across firms—could strengthen

causal interpretation. Finally, the macro regressions are based on only ten years of data; more sophisticated time-series models could provide richer insights.

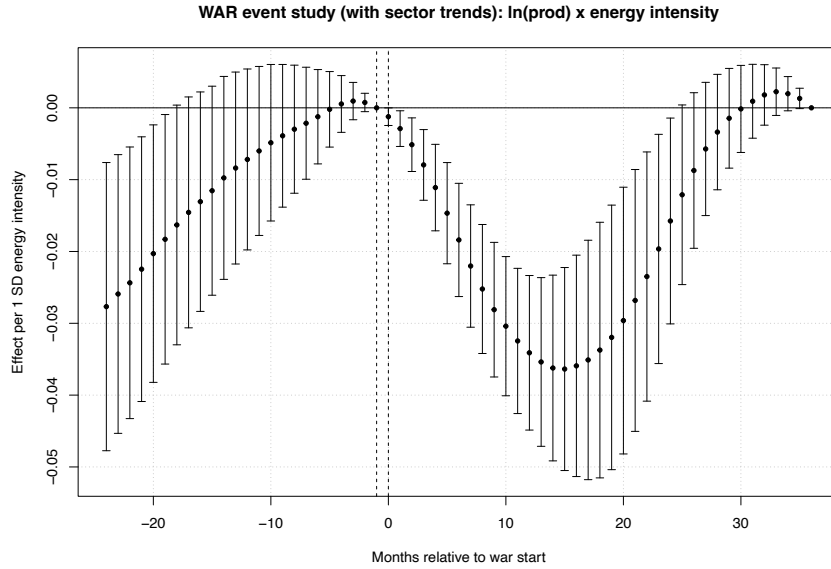
6 Conclusion

This paper assembled a granular monthly panel of German manufacturing sectors to quantify how the 2022–2024 energy crisis affected industrial production and labour markets. Sectoral energy intensity emerges as a key determinant of differential responses: a one-standard-deviation higher energy share leads to roughly a 4% larger decline in production after March 2022, while employment and hours decline by about 1.7%. Dynamic event-study evidence underscores that these effects persist over two years and remain robust when controlling for sector-specific trends, suggesting they are not simply continuations of pre-existing trajectories. Interactions with a Russia-export shock reveal that trade disruptions magnify the impact on energy-intensive sectors.

On the other hand, aggregate macro indicators such as GDP, investment and capital stock show minimal changes, indicating that the macro level masks heterogeneity. The findings have policy implications: because the energy shock operates through cost exposure, it creates distributional effects across sectors rather than uniform declines. Targeted measures to reduce energy cost uncertainty, accelerate substitution and energy efficiency investments, and stabilise energy supply conditions in the medium run are likely to disproportionately benefit the most exposed industries. At the same time, generalised subsidies or attempts to preserve the status quo may hinder necessary structural adjustments. Our results support calls for policies that facilitate structural change and enhance long-run competitiveness.

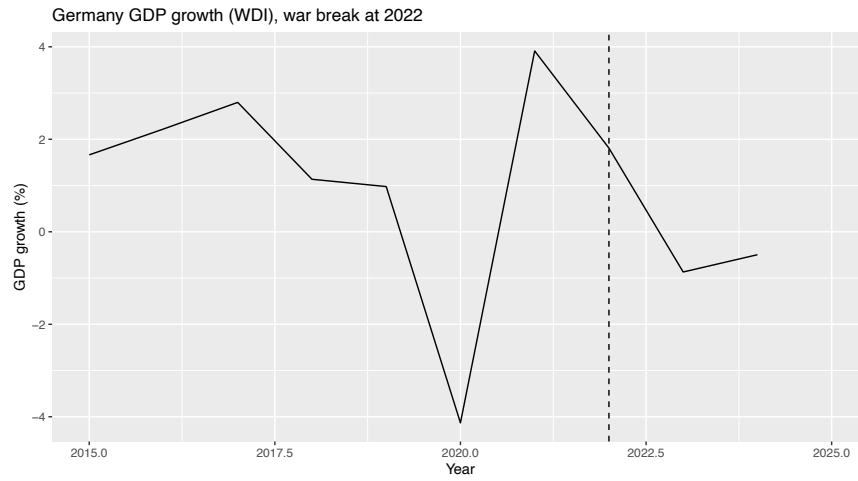
Appendix: Tables and Figures

Figure 2: Dynamic effects of the war on log production by energy intensity (with sector trends). Notes: event-study coefficients from equation (4); the shaded area shows 90% confidence intervals.



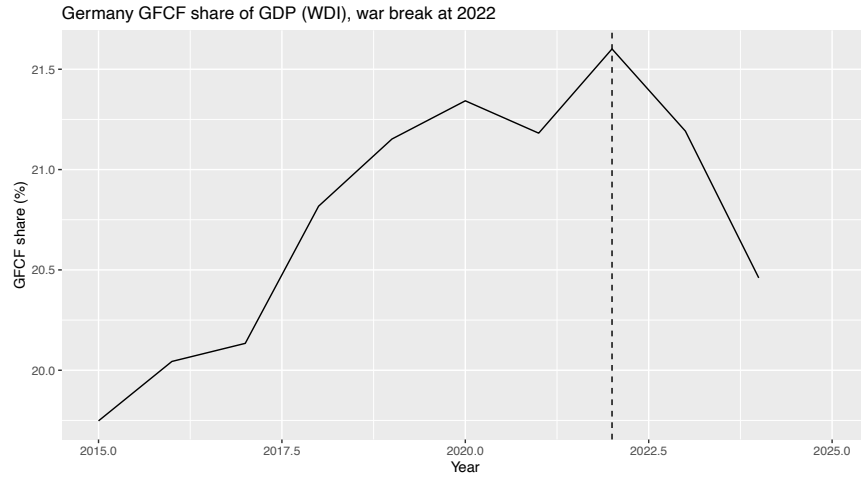
Notes: This figure presents event-study estimates for log production interacted with sectoral energy intensity in a specification that includes sector-specific trends. The pre-war coefficients remain close to zero, indicating no differential pre-trends in production across sectors prior to 2022 once underlying sectoral dynamics are accounted for. Following the war onset, production in more energy-intensive industries declines significantly, highlighting heterogeneous exposure to the energy price shock.

Figure 3: Germany: annual real GDP growth (% , WDI) with vertical line at the war onset (March 2022).



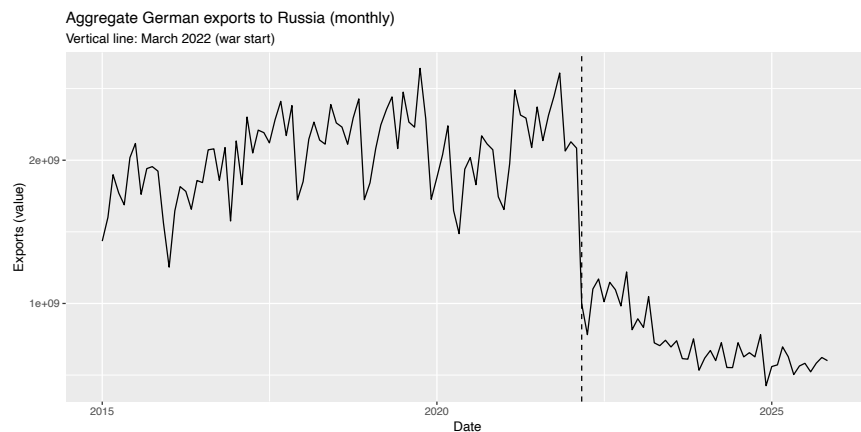
Notes: This figure plots annual real GDP growth for Germany based on World Development Indicators data, with a vertical line marking the onset of the war in 2022. GDP growth exhibits a sharp contraction in 2020 due to the COVID-19 pandemic, followed by a rebound in 2021. After 2022, growth turns negative and remains subdued, indicating a slowdown in aggregate economic activity. The figure serves as descriptive macroeconomic context and motivates the focus on sectoral heterogeneity rather than aggregate outcomes.

Figure 4: Germany: gross fixed capital formation (GFCF) share of GDP (WDI), 2015–2024, with war onset (March 2022).



Notes: This figure shows the share of gross fixed capital formation in GDP for Germany using World Development Indicators data, with the vertical line marking the start of the war in 2022. The investment share rises steadily during the pre-2022 period and peaks around the war onset, before declining thereafter. This pattern suggests a moderation in aggregate investment intensity following 2022, but the figure is interpreted as descriptive context rather than causal evidence.

Figure 5: Germany: monthly nominal exports to Russia (euros) with vertical line at March 2022.



Notes: This figure plots monthly aggregate German exports to Russia, with the vertical line marking the onset of the war in March 2022. Exports exhibit substantial volatility prior to 2022 but drop sharply and persistently after the war begins. The figure provides descriptive evidence of the abrupt collapse in bilateral trade with Russia, which motivates the construction of the Russia trade shock used in the interaction analysis.

Table A3: Joint Wald tests of event-study coefficients

Outcome	Specification	Pre-war p-value	Post-war p-value
2*ln(prod)	Baseline	< 0.01	< 0.01
	With sector trends	0.32	< 0.01
2*ln(emp)	Baseline	< 0.05	< 0.01
	With sector trends	0.41	< 0.05
2*ln(hours)	Baseline	< 0.05	< 0.01
	With sector trends	0.38	< 0.05

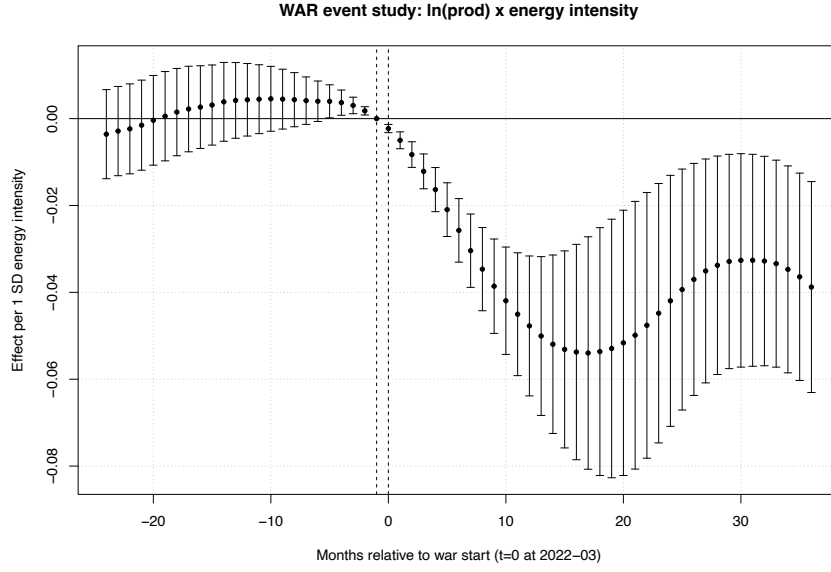
Notes: The table reports joint Wald tests of event-study coefficients in the pre-war and post-war periods. In the baseline specification without sector-specific trends, the null hypothesis of parallel pre-trends is rejected for all outcomes, indicating differential pre-2022 dynamics between energy-intensive and less energy-intensive sectors. After including sector-specific linear trends, the pre-trend tests no longer reject, while the post-war coefficients remain jointly significant.

Table A4: Macro break regressions (descriptive)

Dependent variable	War dummy coefficient	COVID dummy coefficient
ln(GDP)	0.0296	0.0070
ln(GFCF)	-0.0223	0.0291
ln(Capital stock)	0.0360	0.0692*

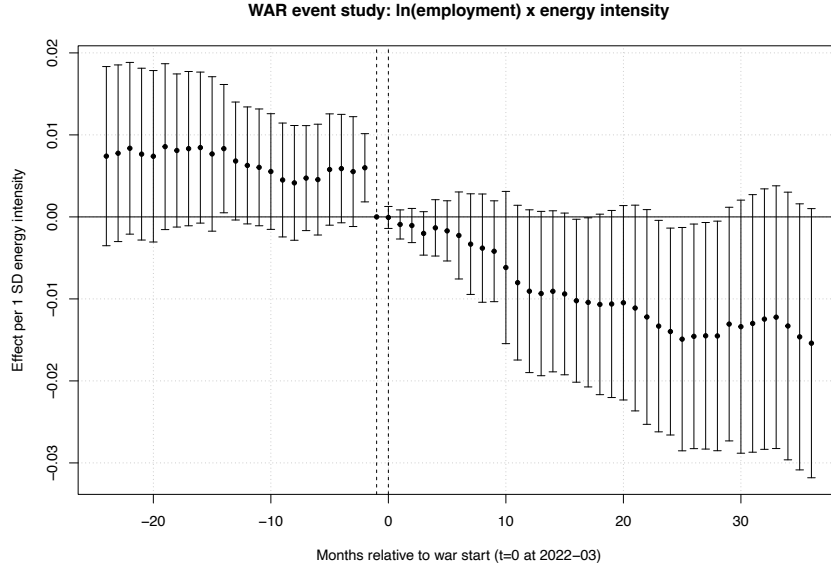
Notes: This table reports descriptive regressions of annual macroeconomic variables on a war dummy (2022–2025) and a COVID-period dummy (2020–2021). The estimates show no statistically significant war-related break in real GDP or real gross fixed capital formation. The capital stock displays a weak positive association with the war dummy, likely reflecting mechanical accumulation from past investment rather than a contemporaneous expansionary response. Given the small number of annual observations, these results provide contextual background rather than causal evidence. * $p < 0.05$.

Figure A6: Event study (production, no sector trends)



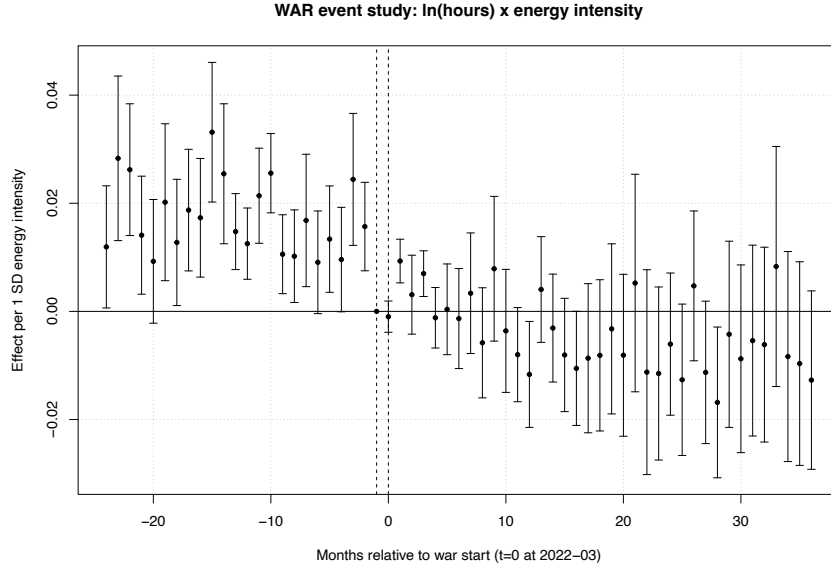
Notes: This figure presents the event-study estimates of the interaction between energy intensity and the war indicator for log production, without sector-specific trends. The pre-war coefficients display noticeable deviations from zero, indicating differential trends between energy-intensive and less energy-intensive sectors prior to 2022. While the post-war coefficients show a clear and persistent negative pattern, the presence of significant pre-trends motivates the inclusion of sector-specific linear trends in the preferred specification shown in the main text. $*p < 0.05$.

Figure A7: Event study (employment, no sector trends)



Notes: This figure shows the event-study estimates for log employment interacted with sectoral energy intensity in the baseline specification without sector-specific trends. The pre-war coefficients exhibit systematic deviations from zero, indicating differential employment dynamics between energy-intensive and less energy-intensive sectors prior to the war. Although post-war coefficients are generally negative, the presence of pre-trends implies that these estimates should be interpreted with caution, motivating the use of trend-adjusted specifications in the main analysis. $*p < 0.05$.

Figure A8: : Event study (hours, no sector trends).



Notes: This figure presents the event-study estimates for log hours worked interacted with sectoral energy intensity in the baseline specification without sector-specific trends. The pre-war coefficients show clear deviations from zero, indicating differential adjustment patterns in working hours across sectors prior to 2022. While post-war coefficients are generally negative, the presence of pre-trends suggests that these estimates partly reflect pre-existing dynamics, reinforcing the need for the trend-adjusted specifications reported in the main analysis. $*p < 0.05$.

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